

Quaternionic Logarithm

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Abstract

We examine the analytical structure of the quaternionic logarithm and introduce the concepts of linear integration in quaternionic space, coherent with that of the regular complex analysis. Along the way, we bring about the suitable definitions of quaternionic derivatives, leaving aside the longstanding problem of quaternionic analyticity. We briefly discuss possible implications of this analysis for physical scenarios

We also set the goal of analysis various types of singularities encountered in physics from the quaternionic perspective

1 Gateway

A typical lecture on introductory mathematical physics often includes a pedagogical derivation of the residue of a holomorphic function,

$$\oint f(z) dz = 2\pi i \cdot \text{Res } f(0).$$

This calculation would typically be done in an explicit approach, by expanding $f(z)$ into a series (assuming it is expandable),

$$f(z) = \dots + \frac{a_{-2}}{z^2} + \frac{a_{-1}}{z} + a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots,$$

choosing a circular contour around the origin and parametrising it by an angular variable φ . This way,

$$\begin{aligned} z^n &= e^{in\varphi}, \\ dz &= i z(\varphi) d\varphi. \end{aligned}$$

Then it turns out that the integral of any power of z vanishes, as z is periodic in φ ,

$$i a_n \int_0^{2\pi} z^{n+1}(\varphi) d\varphi = \frac{a_n}{n+1} z^{n+1}(\varphi) \Big|_0^{2\pi} = 0.$$

That is — except when $n = -1$ — in which case the integrand is a constant. This constant then integrates to the celebrated value of $2\pi i$:

$$a_{-1} \oint \frac{dz}{z} = i a_{-1} \int_0^{2\pi} d\varphi = 2\pi i \cdot a_{-1}.$$

What really happens, and what is easier to explain — but what is not typically done during such an introductory lecture because of the lack of the mathematical basis, which is only being introduced — is the following. Let us take a naïve integral of any integer power of z , the same way as we would compute one of a real function of x ,

$$\int z^n dz = \frac{1}{n+1} z^{n+1}.$$

Now, when we evaluate z at the beginning and at the end of the contour, we realise that we have to “calculate” z^{n+1} at one at the same point (say, $z = 1$). And so the

integral vanishes. But when we pick the power $1/z$, this formula no longer works — the integral no longer yields an integer power of z . Instead, it gives us the logarithm,

$$\int \frac{dz}{z} = \log z.$$

At this point we realise, that the logarithm is *not* evaluated at one and the same point. It is instead evaluated at the same position on two different branches of the Riemann surface of the logarithm. Hence the $2\pi i$.

We now re-phrase this conclusion, to emphasize it — the reason that the integral of dz/z around a closed contour, has given a non-zero result (in contrast to all other powers of z), is that the contour in fact is *not closed*

A naïve student, beginning to study Complex Analysis, would think, that an integral of a (smooth) function should not depend on the *path* of the integration. Otherwise, it is not quite clear how to compute integrals. And therefore an integral of any function around a closed contour should vanish. And we can see that such naïve thinking, applied to holomorphic functions, almost works, except for the case of a prime pole, when the contour, strictly speaking, is not closed

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Of course, everything that we have just described has been known for almost two hundred years. However, one can still ask some quite naïve questions. We can state, that the integral of any holomorphic function around a closed contour vanishes, provided that the contour is in fact closed. The reason that functions have residues, is that when we try to encircle their poles by a contour, we end up with a non-closed contour. The contour

around the origin does not end at the same point where it began (we will talk about why, in a moment).

But is it possible to *close* this contour, such that the integral along it *would* vanish?

Now, indeed, why do we say that the contour around the origin is not closed? To this end, function $f(z) = 1/z$ is smooth and single-valued, it does *not* have any branch

cuts. Yet, when we calculate its integral we end up with a multi-branch function. How does that happen?

Well, there is nothing mysterious. When we say that we “calculate” the integral

$$\int z^n dz ,$$

we in fact imply that the 1-form $z^n dz$ is exact, and,

$$z^n dz = \frac{1}{n+1} d z^{n+1} .$$

Then, applying integration to both sides, we end up with the function z^{n+1} . This argument does not pass through with the function $1/z$. The 1-form dz/z is *closed*, but not *exact*,

$$\frac{dz}{z} \neq d \log z .$$

Locally, these two forms are equal everywhere (except for the branch cut), but not globally. They cannot possibly be equal globally, just for the fact that the logarithm is a multi-valued function, and has a branch cut (which, although not a singularity of course, in this comparison can be viewed as a “discontinuity”). Function dz/z , in contrast, is perfectly smooth and defined everywhere on the normal complex plane, just less the origin.

This means, that the form dz/z in the equality

$$\int \frac{dz}{z} = \log z , \tag{1.1}$$

is not entirely the same as one would normally think, but rather, its extension to the Riemann surface of the logarithm. With this clarification, the equality (1.1) is correct. So, we have had to *extend* function $1/z$ to the Riemann surface of $\log z$, and only then could we perform the integration.

The following questions arise,

did function $1/z$ in fact know that its integral would be multi-valued?

did it have that information encoded anywhere?

It turns out that yes, in a way, it did know that. As for question (??), it *is* possible to close the contour γ in such a way, that the integral around it would vanish. In order to do that, however, we need to step out of the complex plane. On that path, we realise that we are entering a whole new world.

2 Analyticity

In this section we explain why we do *not* address the issue of quaternionic analyticity in this work. The issue becomes apparent during the variable counting

Consider arbitrary complex functions $f(z, \bar{z})$. These consist of two real functions of two real variables. In terms of the real-valued analysis, a derivative of a complex function is a 2×2 symmetric real matrix,

$$\partial f(z, \bar{z}) = \partial(u + iv) = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}.$$

This derivative can also as well be described in terms of two complex derivatives — $\partial_z f$ and $\partial_{\bar{z}} f$

The complex analyticity condition

$$\partial_{\bar{z}} f(z, \bar{z}) = 0, \tag{2.1}$$

imposes two real conditions, leaving us with two independent real functions, which form a single, holomorphic function. It makes sense to denote such a function as $f(z)$ because condition (2.1) implies that this function is independent of $\bar{z}$ as of a linear combination $x - iy$. This condition also implies that integration of $f(z)$ can be written and performed in the conventional real-analysis fashion,

$$f(b) - f(a) = \int_a^b \partial f = \int_a^b \partial_z f(z) dz.$$

In contrast, a general quaternionic function consists of four functions of four real variables. Therefore, a derivative of such a function is a 4×4 symmetric matrix

$$\partial f(q) = \begin{pmatrix} \partial_t f_0 & \partial_x f_0 & \dots & \partial_z f_1 \\ & \dots & \dots & \\ & \dots & \dots & \\ \partial_t f_3 & \dots & \dots & \partial_z f_3 \end{pmatrix},$$

which would require *four* quaternionic derivative functions for an equivalent description. And this is what we will have to do — we will introduce four linearly independent quaternionic derivatives.

But the real problem now is that, unlike in the complex case, one cannot require “analytical” functions to be independent from any linear combination other than $t + \hat{i}x + \hat{j}y + \hat{k}z$. This is because variables t, x, y and z can each be represented as an “analytical function” of q . And therefore, any smooth four-component function of real variables t, x, y and z must be considered an analytical quaternionic function. Otherwise the analyticity conditions become too restrictive.

This does not preclude certain choices of “preferred” conditions of analyticity, as we will discuss below, but they are still too restrictive in our opinion.

The problem now is that it is almost meaningless to perform integration of a quaternionic function along a path by itself. While this would lead to a certain numerical or algebraic result, it would not be related to what we wish to call an integral of a function. In other words,

$$\partial f(q) dq \neq df(q).$$

Instead, all four linearly independent quaternionic derivatives would have to be involved in order to find the “change” of the antiderivative as the result,

$$df(q) = \frac{\partial f(q) dq + \partial_{(\hat{i})} f(q) dq^{(\hat{i})} + \partial_{(\hat{j})} f(q) dq^{(\hat{j})} + \partial_{(\hat{k})} f(q) dq^{(\hat{k})}}{4}.$$

We call such derivatives *sister* derivatives to the original derivative $\partial f(q)$, and introduce them in the Section 3.

In other words, all four sister derivatives need to be known in order to perform quaternionic integration. Or, in general, one needs four functions in order to perform integration along a contour.

2.1 Pure functions

How does this help us with integrating our function $1/z$? This function is a derivative of $\log z$, but do we know its sister derivatives?

Before addressing these questions let us introduce a concept of *pure* functions. Any algebraic complex analytical function $f(z)$ can be easily promoted to a quaternionic function $f(q)$:

$$f(z) = e^z \quad \longrightarrow \quad f(q) = e^q.$$

We call such functions *pure*, because they do not involve fixed quaternionic constants. A more precise definition can be done, but is not necessary for what follows. It is obvious that any analytical complex function can be promoted this way,

$$\sin z \longrightarrow \sin q, \qquad \log z \longrightarrow \log q.$$

While such functions not necessarily are useful for the quaternionic analysis that we are going to dive into, they are useful in relating the quaternionic properties that we find for them to their complex counterparts

TODO: discuss the mapping of i

3 Conjugations

We need to elaborate on the concept of the quaternionic conjugation and introduce the relevant notations. Given a quaternion

$$q = t + \hat{i}x + \hat{j}y + \hat{k}z,$$

we denote the usual quaternionic conjugate of q as

$$\tilde{q} = t - \hat{i}x - \hat{j}y - \hat{k}z. \quad (3.1)$$

We then introduce directional quaternionic conjugates,

$$\begin{aligned} q^{(\hat{i})} &= -\hat{i}q\hat{i} = t + \hat{i}x - \hat{j}y - \hat{k}z, \\ q^{(\hat{j})} &= -\hat{j}q\hat{j} = t - \hat{i}x + \hat{j}y - \hat{k}z, \\ q^{(\hat{k})} &= -\hat{k}q\hat{k} = t - \hat{i}x - \hat{j}y + \hat{k}z, \end{aligned} \quad (3.2)$$

and in general, for any unit vector $\hat{a}$,

$$q^{(\hat{a})} = -\hat{a}q\hat{a}. \quad (3.3)$$

The composition of the conjugations (3.2) can be readily found,

$$(q^{(\hat{i})})^{(\hat{j})} = (q^{(\hat{j})})^{(\hat{i})} = q^{(\hat{k})}, \quad (q^{(\hat{i})})^{(\hat{i})} = q.$$

These are valid of course as long as $\hat{i}$, $\hat{j}$ and $\hat{k}$ are all orthogonal

We notice that the conjugations (3.2) always flip *two* signs. In contrast, any composition with the regular quaternionic conjugation (3.1) will flip an odd number of signs,

$$\widetilde{(q^{(\hat{i})})} = t - \hat{i}x + \hat{j}y + \hat{k}z.$$

We observe that the conjugations (3.2) and (3.1) are all completely interchangeable,

$$\widetilde{(q^{(\hat{i})})} = (\tilde{q})^{(\hat{i})}.$$

The following obvious identities can be easily inferred,

$$\begin{aligned}(a b)^{(i)} &= a^{(i)} b^{(i)}, \\ (a \tilde{b})^{(i)} &= \tilde{b}^{(i)} \tilde{a}^{(i)}, \\ a^{(i)} b &= (a b^{(i)})^{(i)}.\end{aligned}$$

We can also use the directed conjugations to isolate any component of q ,

$$\begin{aligned}t &= \frac{1}{4} \left\{ q + q^{(i)} + q^{(j)} + q^{(\hat{k})} \right\}, \\ x &= \frac{1}{4 \hat{i}} \left\{ q + q^{(i)} - q^{(j)} - q^{(\hat{k})} \right\}, \\ y &= \frac{1}{4 \hat{j}} \left\{ q - q^{(i)} + q^{(j)} - q^{(\hat{k})} \right\}, \\ z &= \frac{1}{4 \hat{k}} \left\{ q - q^{(i)} - q^{(j)} + q^{(\hat{k})} \right\},\end{aligned}\tag{3.4}$$

or even express $\tilde{q}$ in terms of them,

$$\tilde{q} = \frac{1}{4} \left\{ q - q^{(i)} - q^{(j)} - q^{(\hat{k})} \right\}.$$

While being very trivial, identities (3.2) and (3.4) make it especially obvious, that any real-analytical function of coordinates t , x , y and z is essentially an analytical function of quaternions, and it is not possible to demand a function to only depend on a certain linear combination of these coordinates (which is what is done in the definition of complex analyticity)

The following quite trivial identity will be found useful,

$$\operatorname{Re} q = \operatorname{Re} \tilde{q} = \operatorname{Re} q^{(i)} = \operatorname{Re} q^{(j)} = \operatorname{Re} q^{(\hat{k})}.$$

3.1 Scalar Product

For what follows, it is necessary to introduce the notations for the “scalar product” for quaternions. We will be denoting the scalar product both for quaternions and for 3-vectors using angle brackets,

$$\begin{aligned}\langle a, b \rangle &= a_\mu b^\mu = a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3, \\ \langle \vec{a}, \vec{b} \rangle &= a^1 b^1 + a^2 b^2 + a^3 b^3,\end{aligned}$$

where we now have used the usual index notations for the components of 3- and 4-vectors

It is trivial to express the scalar product of two quaternions in terms of their conjugates,

$$\langle a, b \rangle = \frac{1}{2} \left\{ \tilde{a} b + \tilde{b} a \right\} = \frac{1}{2} \left\{ a \tilde{b} + b \tilde{a} \right\}.$$

Essentially, the usual quaternionic conjugation turns a contravariant 4-vector into a covariant one

Similarly, and suddenly more interestingly, the directed conjugates can be used,

$$\begin{aligned} \langle a, b \rangle &= \frac{1}{4} \left\{ \tilde{a} b + \tilde{a}^{(i)} b^{(i)} + \tilde{a}^{(j)} b^{(j)} + \tilde{a}^{(k)} b^{(k)} \right\} \\ &= \frac{1}{4} \left\{ a \tilde{b} + a^{(i)} \tilde{b}^{(i)} + a^{(j)} \tilde{b}^{(j)} + a^{(k)} \tilde{b}^{(k)} \right\}. \end{aligned} \quad (3.5)$$

These expressions are invariant under any of the conjugations that we have introduced, as well as under replacing $a \leftrightarrow b$

However, if we now introduce the greek-letter notation for the directed conjugates,

$$q^{(\mu)} : \quad q^{(0)} \equiv q, \quad q^{(1)} \equiv q^{(i)}, \quad q^{(2)} \equiv q^{(j)}, \quad q^{(3)} \equiv q^{(k)},$$

the scalar product can suddenly be written in a compact way,

$$\langle a, b \rangle = \frac{1}{4} \tilde{a}_{(\mu)} b^{(\mu)} = \frac{1}{4} a^{(\mu)} \tilde{b}_{(\mu)}. \quad (3.6)$$

Since the expression on the left-hand side is real, the expressions on the right-hand side do not depend on the order of a and b , despite that all of them are, in fact, quaternions.

Expression (3.6) is now, unexpectedly, very similar to the usual expression for the scalar product $a_\mu b^\mu$. We chose to lower index (μ) on $\tilde{b}$ just for convenience and for this resemblance. Despite that $q^{(\mu)}$ looks like “components” of vector q , they are, of course, not its components, nor are they even real.

However, it is because of this resemblance that we will be able to manage quaternionic expressions involved into integration and differentiation

4 Differential

5 Derivatives

6 Logarithm

7 Integration

8 Conclusions

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A Spinors

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